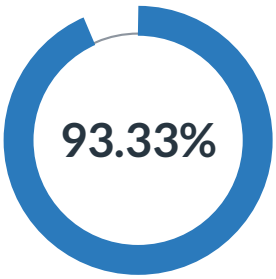
Results

Carter Feng



42
Out of 45 points

37:34
Time for this attempt

Your Answers:

1 0 / 0 points Fill in the Blank

At Seven Lakes High School, we believe that character matters and that academic integrity reflects who we are as learners and citizens. By signing this pledge, I commit to upholding the values of honesty, responsibility, and respect in all academic and personal endeavors.

My Academic Integrity Pledge:

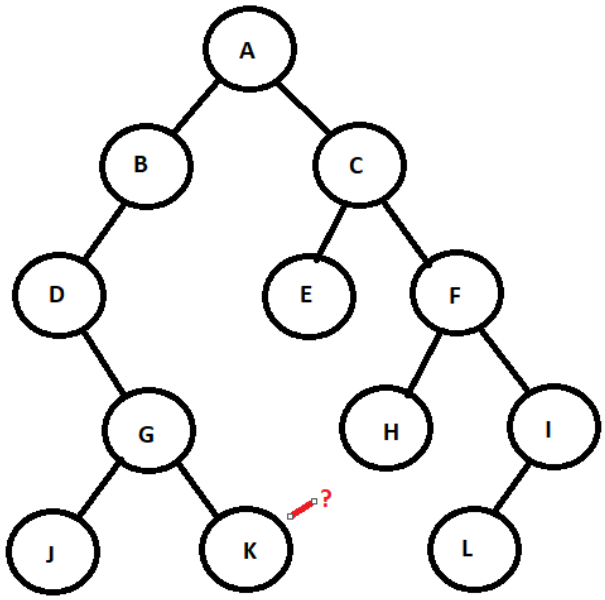
- **I will complete all academic work honestly, without giving or receiving unauthorized assistance.**
- **I will not copy, plagiarize, or submit someone else's work as my own.**
- **I will not use unauthorized materials (notes, devices, online resources) during assessments.**
- **I will respect the integrity of others' work and will not aid others in academic dishonesty.**
- **I will not share or receive academic work, ideas, or answers electronically or in-person when individual work is expected or collaboration is not permitted.**
- **I will report any knowledge of cheating or violations of academic integrity to a trusted adult.**

I understand that academic dishonesty harms my character, undermines trust in our community, and may result in consequences outlined in the school's academic integrity policy.

Print your name to acknowledge the oath. Not printing your name will result in a zero for the assessment.

✓ Carter Feng

2 1 / 1 point Multiple Answer



Identify the marked item [?].

- [ ] parent, internal node, inner node, branch node
- [x] sibling
- [ ] left child
- [x] right child
- [ ] root
- [x] leaf, outer node, terminal node
- [ ] path
- [ ] edge

3 1 / 1 point Multiple Choice

What is the worst case big O time constraint for searching for a no

- $O(N^2)$
- $O(\log N)$
- $O(N \log N)$
- ✓ **$O(N)$**
- $O(1)$

4 1 / 1 point Multiple Choice

What is a complete tree?

- A tree whose only have nodes of degree 0 and degree 2
- A tree whose leaves are at the same depth.
- A tree that whose left subtree and right subtree heights differ by no more than 1 level
- ✓ **A tree whose levels are at maximum width, with the exception of the lowest level where all the nodes in that level are as far left as possible**

5 1 / 1 point Multiple Choice

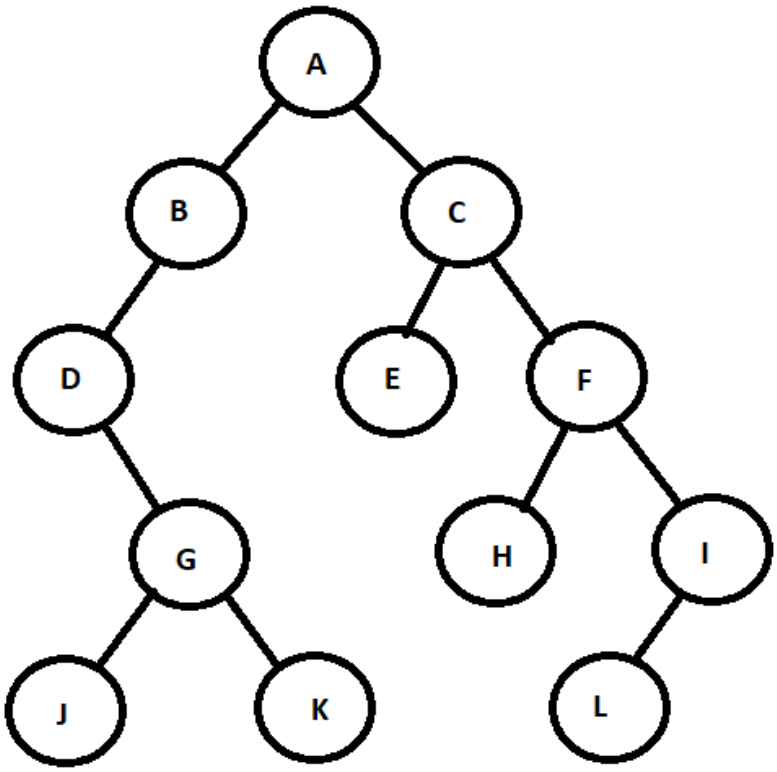
What is a full tree?

- A tree whose leaves are at the same depth.
- A tree that whose left subtree and right subtree heights differ by no more than 1 level

A tree whose levels are at maximum width, with the exception of the lowest level where all the nodes in that level are as far left as possible

✓ A tree whose only have nodes of degree 0 and degree 2

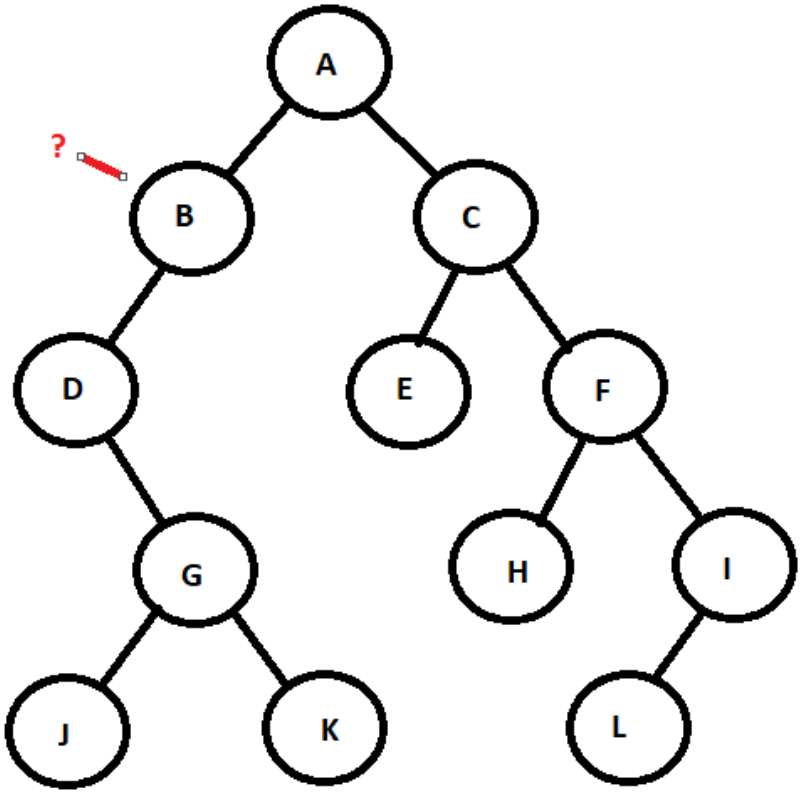
6 1 / 1 point Fill in the Blank



What is the level location of node J

✓ 4

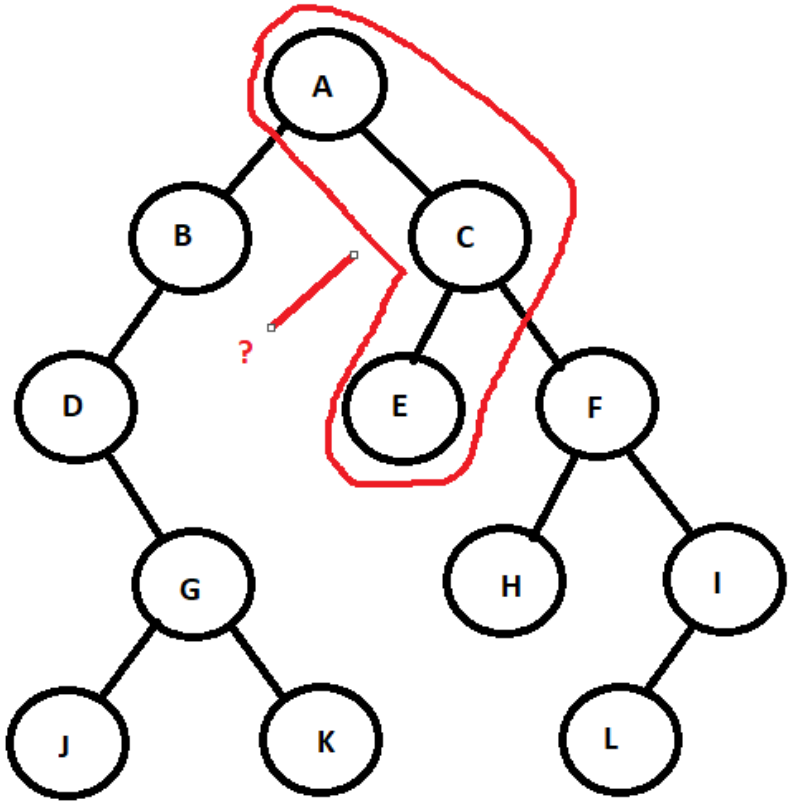
7 1 / 1 point Multiple Answer



Identify the marked item [?].

- [x] sibling
- [ ] edge
- [ ] right child
- [ ] root
- [ ] path
- [x] parent, internal node, inner node, branch node
- [x] left child
- [ ] leaf, outer node, terminal node

8 0 / 1 point Multiple Answer



Identify the marked item [?].

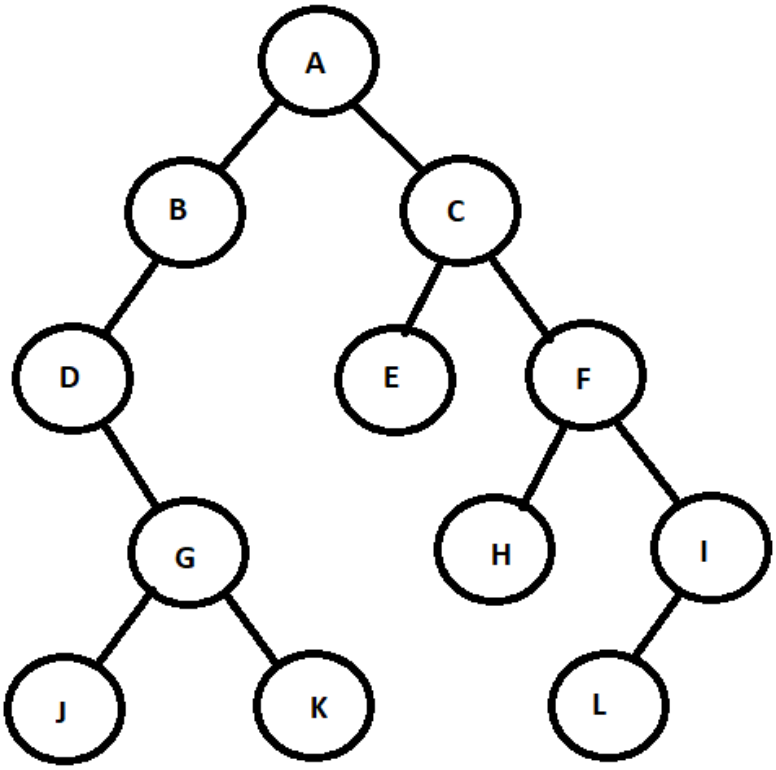
- [ ] leaf, outer node, terminal node
- [x] branch, subtree

Selected Answer - Incorrect

- [ ] root
- [ ] left child
- [ ] right child
- [ ] sibling
- [ ] edge
- [ ] parent, internal node, inner node, branch node
- [ ] path

Missed Option - Incorrect

9 1 / 1 point Fill in the Blank



What is the level location of node B

✓ 1

10 0 / 1 point Multiple Choice

What is the big O time constraint for adding a value to a binary search tree?

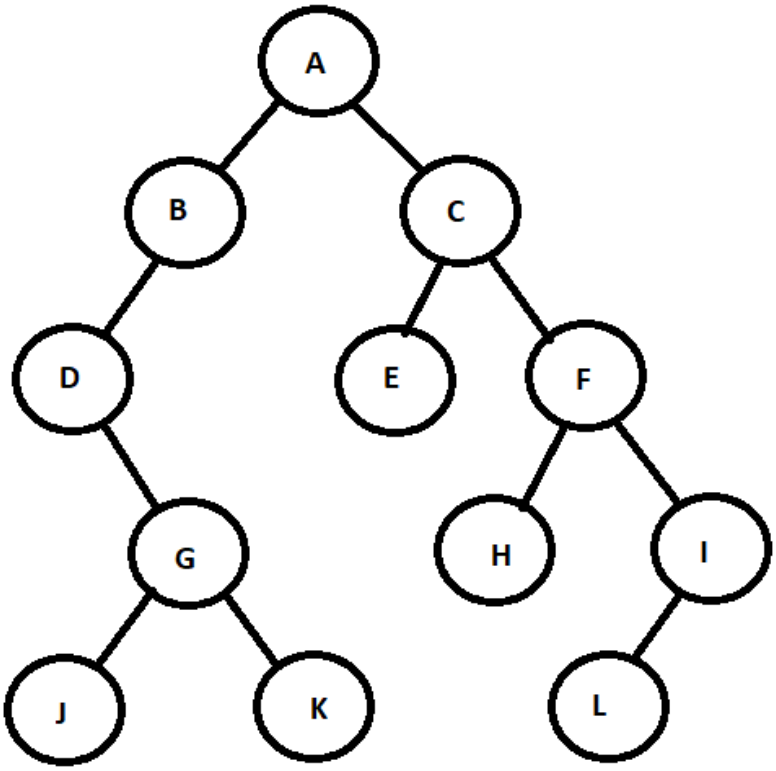
- O(N log N)
- $O(N^2)$
- ✓ O(N)

Correct Answer: **O(log N)**

O(1)

O(log N)

11 1 / 1 point Fill in the Blank



What is the width of this particular tree?

✓ 3

12 30 / 30 points Matching

The object of this question is to build a binary search tree from scratch. The programmer will be showing what the binary search tree looks like after certain values are added. Here is the first set of numbers to add

19, 5, 27, 44, 34, 40, 14, 38, 49, 17

What is the height of the tree after the value 17 is entered? [A]

What is the width of the tree after the value 17 is entered? [B]

What is the preorder traversal after the value 17 is entered?

[C0]	[C1]	[C2]	[C3]	[C4]	[C5]	[C6]	[C7]	[C8]	[C9]

Add the following to tree 25, 22, 26, 50, 2, 10, 12, 4, 41, 6

What is the diameter of the tree after the value 6 is entered? [D]

How many nodes of degree 0 are there in the tree after the value 6 is entered? [E]

What are the first 10 values printed in a postorder traversal after the value 6 is entered?

[F0]	[F1]	[F2]	[F3]	[F4]	[F5]	[F6]	[F7]	[F8]	[F9]

How many degree 2 nodes are there? [G]

Whati si the width of level 4 of the tree? [H]

Remove the following values 19, 14, 40, 17, 2.

What are the values, from left to right, on level 3 of the tree?

If there is no node in a particular spot, write null.

[I0]	[I1]	[I2]	[I3]	[I4]	[I5]	[I6]	[I7]

F5	✓ 17
F1	✓ 2
I6	✓ 34
I7	✓ 49
F0	✓ 4
C9	✓ 49
C8	✓ 38
H	✓ 4
C1	✓ 5
E	✓ 9
C7	✓ 40

F4	✓	10
G	✓	8
F7	✓	5
B	✓	3
F3	✓	12
F6	✓	14
C6	✓	34
C0	✓	19
F9	✓	26
I3	✓	12
C5	✓	44
I4	✓	null
A	✓	5

C3	✓	17
F8	✓	22
I5	✓	26
C4	✓	27
F2	✓	6
I0	✓	null
I1	✓	null
C2	✓	14
I2	✓	6
D	✓	10

13 4 / 5 points Essay

Write the method diameter()

The method diameter returns the diameter of the tree that runs through the root.

When writing diameter, assume that the BinarySearchTree has a method called getHeight(BinaryNode k) which will give the height of a subtree whose root is k.

public int diameter()
{

```
public int diameter() {
  if(root==null) {
    return;
  }
  if(root.left()!=null) {
    int left = getHeight(root.left()) + 1;
  } else {
    int left = 0;
  }
  if(root.right()!=null) {
    int right = getHeight(root.right()) + 1;
  } else {
    int right = 0;
  }
  return left + right + 1;
}
```

Graded